Geometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 5$ in solving contextual problems with integers, evaluate $f(5)$.

Final Answer: 25. Substitute $x = 5$: $f(5) = 4(5) + 5 = 25$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 6$ in fractions and the number line, evaluate $f(2)$.

Final Answer: 16. Substitute $x = 2$: $f(2) = 5(2) + 6 = 16$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for fractions and the number line consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 18.5. The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 7$ in multiplying and dividing integers, evaluate $f(2)$.

Final Answer: 19. Substitute $x = 2$: $f(2) = 6(2) + 7 = 19$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for multiplying and dividing integers consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 19.5. The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 2x + 8$ in mixed operations with integers, evaluate $f(2)$.

Final Answer: 12. Substitute $x = 2$: $f(2) = 2(2) + 8 = 12$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for mixed operations with integers consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 20.5. The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Segment AB has endpoints $A(2,\ -1)$ and $B(6,7)$. Find the equation of the perpendicular bisector of AB in slope-intercept form.

Solution: The perpendicular bisector passes through the midpoint and has a perpendicular slope. Step 1 (Midpoint): $M = \left(\frac{2+6}{2},\ \frac{-1+7}{2}\right) = (4,3)$. Step 2 (Slope of AB): $m_{AB} = \frac{7-(-1)}{6-2} = \frac{8}{4} = 2$. Step 3 (Perpendicular slope): $m_{\perp} = -\frac{1}{2}$. Step 4 (Point-slope equation through M(4, 3)): $y-3 = -\frac{1}{2}(x-4) \Rightarrow y-3 = -\frac{1}{2}x+2 \Rightarrow y = -\frac{1}{2}x+5$.

9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Find the equation of the perpendicular bisector of the line segment joining $A(-2,5)$ and $B(4,\ -3)$. Give your answer in the form $ax+by+c=0$, where a, b, c are integers.

1. Midpoint of AB: $M = \left(\frac{-2+4}{2}, \frac{5-3}{2}\right) = (1,1)$. 2. Gradient of AB: $m_{AB} = \frac{-3-5}{4-(-2)} = \frac{-8}{6} = -\frac{4}{3}$. 3. Gradient of perpendicular bisector: $m_{\perp} = -\frac{1}{-4/3} = \frac{3}{4}$. 4. Equation of line passing through $M(1,1)$: $y-1 = \frac{3}{4}(x-1) \Rightarrow 4(y-1) = 3(x-1) \Rightarrow 4y-4 = 3x-3$ Rearrange to standard integer form: $3x-4y+1=0$.

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Find the coordinates of the point P on directed line segment from $A(-3,2)$ to $B(7,12)$ that partitions the segment in the ratio $2:3$.

Step 1: Determine the fractional distance: A ratio of $2:3$ means P is $\frac{2}{2+3} = \frac{2}{5}$ of the way from A to B. Step 2: Apply the partition formula: $x_P = x_1 + \frac{2}{5}(x_2 - x_1) = -3 + \frac{2}{5}(7-(-3)) = -3 + \frac{2}{5}(10) = -3+4 = 1.$ $y_P = y_1 + \frac{2}{5}(y_2 - y_1) = 2 + \frac{2}{5}(12-2) = 2 + \frac{2}{5}(10) = 2+4 = 6.$ The partitioning point is $P(1,6)$.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Find the set of values of the constant k for which the equation $kx^2+4x+(k-3)=0$ has real roots.

For real roots, the discriminant $\Delta = b^2-4ac \geq 0$. Here $a=k$, $b=4$, $c=k-3$: $\Delta = 4^2-4(k)(k-3) = 16-4k^2+12k \geq 0$ Divide the entire inequality by -4, reversing the inequality sign: $k^2-3k-4 \leq 0$ Factor the quadratic: $(k-4)(k+1) \leq 0$. The critical values are $k=-1$ and $k=4$. Since the parabola opens upward, the product is ≤ 0 between the roots: $-1 \leq k \leq 4$. Note that if $k=0$, the equation becomes the linear equation $4x-3=0$, which has one r.

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12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find the equation of the normal to the curve $y = \frac{4}{\sqrt{x}} + 2x$ at the point where $x = 4$.

Rewrite in index form: $y = 4x^{-1/2} + 2x$. At $x = 4$: $y = 4(4^{-1/2}) + 2(4) = 4\left(\frac{1}{2}\right) + 8 = 2 + 8 = 10$. Point of contact is $(4, 10)$. Differentiate: $\frac{dy}{dx} = 4\left(-\frac{1}{2}x^{-3/2}\right) + 2 = -2x^{-3/2} + 2 = -\frac{2}{x^{3/2}} + 2$ At $x = 4$: $x^{3/2} = (\sqrt{4})^3 = 8$. Thus: $m_{\tan} = -\frac{2}{8} + 2 = -\frac{1}{4} + 2 = \frac{7}{4}$ The normal has gradient $m_{\text{norm}} = -\frac{1}{7/4} = -\frac{4}{7}$. Using point $(4, 10)$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A circular archery target has three concentric rings of radii 10 cm, 20 cm, and 30 cm. If an arrow hits the target completely at random, what is the exact probability that it lands in the middle ring (between radii 10 cm and 20 cm)?

Step 1: Calculate the total area of the target: The outer circle has radius $R = 30$ cm: $\text{Area}_{\text{total}} = \pi(30^2) = 900\pi$ cm^2. Step 2: Calculate the area of the middle ring (annulus): The middle ring is bounded between $r_1 = 10$ cm and $r_2 = 20$ cm: $\text{Area}_{\text{ring}} = \pi(20^2) - \pi(10^2) = 400\pi - 100\pi = 300\pi$ cm^2. Step 3: Compute the geometric probability.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A rhombus has diagonals of lengths 16 cm and 12 cm. Find the perimeter and area of the rhombus.

Step 1: Use diagonal properties of a rhombus: The diagonals of a rhombus are perpendicular bisectors of each other. They divide the rhombus into four congruent right triangles with legs of lengths $\frac{16}{2} = 8$ cm and $\frac{12}{2} = 6$ cm. Step 2: Find the side length s: Using the Pythagorean theorem: $s = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$ cm. Step 3: Calculate the perimeter: $\text{Perimeter} = 4s = 4(10) = 40$ cm. Step 4: Calculate the area.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Sets & Venn Diagrams

From an external point P, two tangents PA and PB are drawn to a circle with radius $r = 5$ cm. The distance from P to the center of the circle O is 13 cm. Find the length of the tangents PA and PB, and find the perimeter of quadrilateral $PAOB$.

Step 1: Recognize tangent-radius perpendicularity: Tangents are perpendicular to the radius at the point of tangency, so $\angle PAO = \angle PBO = 90^\circ$. ΔPAO is a right triangle with hypotenuse $PO = 13$ and leg $OA = 5$. Step 2: Solve for tangent length using the Pythagorean theorem: $PA = \sqrt{PO^2 - OA^2} = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$ cm. By equal tangent lengths from an external point, $PB = PA = 12$ cm. Step 3: Compute the perimeter of $PAOB$.

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16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Algebra & Graphs

A block of mass $8\ \text{kg}$ rests in limiting equilibrium on a rough plane inclined at 30° to the horizontal. Taking $g = 9.8\ \text{m/s}^2$: (a) calculate the normal reaction R, (b) find the coefficient of friction μ between the block and the plane.

(a) Resolve perpendicular to the inclined plane: $R = mg\cos 30^\circ = 8(9.8)\cos 30^\circ = 78.4 \times \frac{\sqrt{3}}{2} = 39.2\sqrt{3} \approx 67.9\ \text{N (3 s.f.)}$ (b) Resolve parallel to the plane. Since the block is in limiting equilibrium on the verge of sliding down, friction $F = \mu R$ acts up the plane: $F = mg\sin 30^\circ = 8(9.8)\sin 30^\circ = 78.4 \times 0.5 = 39.2\ \text{N}$ Setting $F = \mu R$.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Coordinate Geometry

A bag contains 5 red and 4 green marbles. Three marbles are drawn at random without replacement. Find the probability that: (a) all three are red, (b) at least two are red.

Total marbles = 9. Number of ways to choose 3 marbles = $\binom{9}{3} = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84$. (a) All 3 are red: $\binom{5}{3} = 10$. Probability: $P(\text{all red}) = \frac{10}{84} = \frac{5}{42} \approx 0.119$ (b) 'At least two red' means exactly 2 red and 1 green, or exactly 3 red: Ways for 2 red, 1 green: $\binom{5}{2} \times \binom{4}{1} = 10 \times 4 = 40$. Ways for 3 red: $\binom{5}{3} = 10$. Total favorable outcomes: $40 + 10 = 50$. Probability.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Mensuration

A train accelerates uniformly from rest to $30\ \text{m/s}$ in $20\ \text{s}$, travels at constant speed $30\ \text{m/s}$ for $60\ \text{s}$, and then decelerates uniformly to rest in $15\ \text{s}$. Find: (a) the acceleration in the first phase, (b) the total distance travelled, (c) the average speed for the entire journey.

(a) $a_1 = \frac{v-u}{t} = \frac{30-0}{20} = 1.5\ \text{m/s}^2$. (b) Total distance is the area of the trapezoid on the v-t graph with parallel sides $60\ \text{s}$ and $20 + 60 + 15 = 95\ \text{s}$, and height $30\ \text{m/s}$: $s = \frac{60+95}{2} \times 30 = \frac{155}{2} \times 30 = 155 \times 15 = 2325\ \text{m}$ (c) Average speed.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Trigonometry

A block of mass $4\ \text{kg}$ is pulled along a rough horizontal plane ($\mu = 0.25$) by a pulling force of magnitude $25\ \text{N}$ acting at 30° above the horizontal. Taking $g = 9.8\ \text{m/s}^2$: (a) show that the normal reaction $R = 26.7\ \text{N}$, (b) find the acceleration of the block.

(a) Resolve vertically (equilibrium in vertical direction): $R + 25\sin 30^\circ = mg \Rightarrow R + 25(0.5) = 4(9.8) \Rightarrow R + 12.5 = 39.2 \Rightarrow R = 26.7\ \text{N}$ (b) Limiting friction force opposes motion: $F = \mu R = 0.25 \times 26.7 = 6.675\ \text{N}$ Resolve horizontally using $F_{\text{net}} = ma$: $25\cos 30^\circ - F = ma \Rightarrow 25\left(\frac{\sqrt{3}}{2}\right) - 6.675 = 4a$.

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20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Transformations & Vectors

Two particles P of mass 3 kg and Q of mass 5 kg are connected by a light inextensible string passing over a smooth fixed pulley. The system is held at rest and released. Taking $g = 9.8$ m/s^2: (a) find the acceleration a of the system, (b) calculate the tension T in the string, (c) find the force exerted by the string on the pulley.

(a) Write $F = ma$ for each particle in its direction of motion: For Q (moving downwards): $5g - T = 5a$ For P (moving upwards): $T - 3g = 3a$ Add the equations: $2g = 8a \Rightarrow a = \frac{2(9.8)}{8} = 2.45$ m/s^2 (b) Substitute a into the equation for P: $T = 3g + 3a = 3(9.8 + 2.45) = 3(12.25) = 36.75$ N ≈ 36.8 N (c) The pulley supports two vertical segments of string with tension T, so the total downward force on the pulley is.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 3$ in integers and the number line, evaluate $f(2)$.

a. 7 ✓

b. 9

c. 4

d. 11

Substitute $x = 2$: $f(2) = 2(2) + 3 = 7$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 4$ in adding and subtracting integers, evaluate $f(2)$.

a. 10 ✓

b. 13

c. 6

d. 14

Substitute $x = 2$: $f(2) = 3(2) + 4 = 10$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for integers and the number line consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5 ✓

d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for adding and subtracting integers consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5 **b.** 15.5

c. 16.5 ✓ **d.** 18.5

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A chord of length 24 cm is at a distance of 5 cm from the centre of a circle. What is the radius of the circle?

a. **13 cm** ✓ **b.** 12 cm

c. 14 cm **d.** 15 cm

The perpendicular from the centre bisects the chord: half length = 12 cm. Radius = $\sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$ cm. [M1 A1]

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A tangent TA touches a circle at A. Chord AB makes an angle of 38° with TA. Point C lies on the major arc AB. What is angle ACB?

a. **38°** ✓ **b.** 142°

c. 76° **d.** 52°

By the alternate segment theorem, the angle between the tangent and the chord equals the angle subtended in the alternate segment: $\angle ACB = 38^\circ$. [B1]

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A tangent touches a circle at point A. Chord AB makes an angle of 48° with the tangent. By the alternate segment theorem, what is the angle subtended by chord AB in the alternate segment?

a. **48°** ✓ **b.** 132°

c. 42° **d.** 96°

By the Alternate Segment Theorem, the angle between a tangent and a chord equals the angle subtended by that chord in the alternate segment: 48°. [B1]

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28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In a circle with centre O, point P lies on the circumference. What is the angle between the tangent at P and the radius OP?

a. 45°

b. 60°

c. 90° ✓

d. 180°

A fundamental circle theorem states that the angle between a tangent and the radius drawn to the point of contact is always 90°. [B1]

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Each interior angle of a regular polygon is 156°. How many sides does the polygon have?

a. 12

b. **15** ✓

c. 18

d. 20

Exterior angle = $180^\circ - 156^\circ = 24^\circ$. Number of sides $n = \frac{360^\circ}{24^\circ} = 15$. [M1 for 180 - 156, A1 for 15]

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Triangle ABC is inscribed in a circle such that AB is a diameter. If angle BAC = 34°, find the size of angle ABC.

a. **56°** ✓

b. 34°

c. 66°

d. 46°

Angle in a semicircle is 90° ($\angle ACB = 90^\circ$). $\angle ABC = 180^\circ - 90^\circ - 34^\circ = 56^\circ$. [M1 A1]

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Points A, B, C, D lie on a circle. Chords AC and BD intersect at X. If angle $\angle ABD = 42^\circ$, what is the size of angle $\angle ACD$?

a. 42° ✓ **b.** 48°

c. 84° **d.** 96°

Angles in the same segment subtended by the same arc AD are equal. Therefore, $\angle ACD = \angle ABD = 42^\circ$. [B1]

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Points A, B, and C lie on a circle. AB is a diameter. What is the size of angle $\angle ACB$?

a. 45° **b.** 60°

c. 90° ✓ **d.** 120°

By the circle theorem 'the angle subtended by a diameter in a semicircle is a right angle', $\angle ACB = 90^\circ$. [B1]

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Points A, B, and C lie on a circle. Angle ABC = 47°. What is the reflex angle AOC subtended at centre O?

a. **266°** ✓ **b.** 94°

c. 188° **d.** 274°

Obtuse angle at centre = $2 \times 47^\circ = 94^\circ$. Reflex angle AOC = $360^\circ - 94^\circ = 266^\circ$. [M1 A1]

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In cyclic quadrilateral PQRS, angle PQR = 104°. What is the size of the opposite angle PSR?

a. **76°** ✓ **b.** 104°

c. 86° **d.** 90°

Opposite angles of a cyclic quadrilateral sum to 180°: $\angle PSR = 180^\circ - 104^\circ = 76^\circ$. [B1]

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In a regular polygon, the sum of all interior angles is 1440°. How many sides does this polygon have?

a. **10** ✓ **b.** 8

c. 12 **d.** 9

Sum of interior angles = $(n - 2) \times 180^\circ = 1440^\circ \Rightarrow n - 2 = 8 \Rightarrow n = 10$. [M1 A1]

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a cyclic quadrilateral $ABCD$, angle $A = 75^\circ$. What is the size of the opposite angle C?

a. **105°** ✓ **b.** 75°

c. 127.5° **d.** 90°

Opposite angles of a cyclic quadrilateral sum to 180°: $\angle C = 180^\circ - 75^\circ = 105^\circ$. [B1]

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The interior angle of a regular polygon is 160°. How many sides does the polygon have?

a. **18** ✓ **b.** 16

c. 20 **d.** 15

Exterior angle = $180^\circ - 160^\circ = 20^\circ$. Number of sides = $\frac{360^\circ}{20^\circ} = 18$. [M1 A1]

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Mensuration

What is the size of each interior angle of a regular 5-sided polygon?

a. **108°** ✓ **b.** 72°

c. 123° **d.** 98°

Exterior angle = $\frac{360^\circ}{5} = 72^\circ$. Interior angle = $180^\circ - 72^\circ = 108^\circ$. [M1 A1]

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Trigonometry

An exterior angle of an isosceles triangle is 110°. If the two non-adjacent opposite angles are equal, what is the measure of each of those two angles?

a. **55° ✓**

b. 70°

c. 35°

d. 60°

Exterior angle equals sum of two opposite interior angles: $2x = 110^\circ \Rightarrow x = 55^\circ$. [M1 A1]

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Transformations & Vectors

Surface area of a sphere radius 5:

a. 25π

b. 50π

c. 75π

d. 100π ✓

Applying Geometry principles, $4\pi r^2 = 4\pi(25) = 100\pi$ Therefore, the verified correct option is 100π.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Probability & Statistics

In a circle, chords AB and CD intersect at point P. If angle ACD = 28°, what is the size of angle ABD?

a. **28° ✓**

b. 62°

c. 56°

d. 152°

Angles in the same segment subtended by the same arc are equal: $\angle ABD = \angle ACD = 28^\circ$. [B1]

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 9.01 Functions

In a regular polygon, each exterior angle is 30°. How many sides does the polygon have?

a. 10

b. **12 ✓**

c. 14

d. 15

Number of sides $n = \frac{360^\circ}{\text{exterior angle}} = \frac{360^\circ}{30^\circ} = 12$. [B1]

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Number

Triangle DEF is an enlargement of triangle ABC with scale factor 2.5. If the area of triangle ABC is 12 cm², what is the area of triangle DEF?

a. 75 cm² ✓

b. 30 cm²

c. 60 cm²

d. 150 cm²

Area scale factor = $k^2 = 2.5^2 = 6.25$. Area of DEF = $12 \times 6.25 = 75\ \text{cm}^2$. [M1 A1]

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 10.01 Sets & Venn Diagrams

Points A, B, C lie on a circle with centre O. If angle $\angle ABC = 35^\circ$, what is the angle $\angle AOC$ subtended at the centre?

a. 70° ✓

b. 35°

c. 17.5°

d. 145°

Angle at centre is twice angle at circumference: $\angle AOC = 2 \times 35^\circ = 70^\circ$. [B1]

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Algebra & Graphs

Two similar triangles have areas in the ratio 16 : 49. What is the ratio of their corresponding perimeters?

a. 4 : 7 ✓

b. 16 : 49

c. 2 : 3.5

d. 256 : 2401

Ratio of perimeters (linear ratio) = $\sqrt{\frac{16}{49}} = \frac{4}{7}$, so 4 : 7. [M1 A1]

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